MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 17 MOD 1

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 17 Mod 1 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic
MS3106	Connector, Plug, Electric, Straight, Solder Contacts, AN Type

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Naval Sea Systems Command (Code Ident 53711)

WS 4783	Ignition Element, Electric, Mark 13 Mod 0

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL613460	Ignition Element, Electric, Mark 13 Mod 0
DL534559	Cartridge, Impulse, Mark 17 Mod 1
DL491834	Container, Inner, for Cartridge Impulse, Mark 17 Type
DL491835	Container, Outer, for Cartridges, Power Source

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5	Ammunition and Explosives Ashore
WR-43	Preparation of Quality Assurance Provisions

Code of Federal Regulations

49 CFR 100-199	Transportation

(Copies of specifications, standards, other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Conformance to documents. The Mark 17 Mod 1 impulse cartridge covered by this specification shall be manufactured in accordance with DL534559 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purposes of this specification the Mark 13 Mod 0 electric ignition element (WS 4783), and the cartridge propellant (Drawing 2164784) are considered primary components.

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3.5 **Ignition element.** The ignition elements shall be furnished from lots which were manufactured within the past 12 months. The ignition elements shall be manufactured in accordance with the drawings listed on DL613460, and shall be from lots which were accepted in accordance with the requirements of WS 4783. The ignition element shall be free of the following defects when X-rayed (see 4.6.5.2): missing, defective or improperly assembled components; charges missing or abnormally small; or any other imperfections which could adversely affect performance.

3.6 **Propellant charge.**

3.6.1 **Quantity.** All propellant charges shall be as specified on Drawing 2164784. Charges shall be determined by weight rather than by volume measurement.

3.6.2 **Smokeless powder, handling, and loading.** The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.6.3 **Smokeless powder, condition.** The smokeless powder used shall exhibit no evidence of decomposition (see 4.6.2)

3.7 **Cartridge requirements.**

3.7.1 **Bridge circuit resistance.** The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.6.4), shall be not less than 0.08 ohm nor greater than 0.14 ohm.

3.7.2 **X-ray.** Cartridges shall be free of imperfections when X-rayed (see 4.6.5.1).

3.8 **Ballistic requirements.** Cartridges shall meet the ballistic requirements listed below when subjected to the test prescribed in 4.6.6.

3.8.1 **Chamber pressure.** The upper (U) and lower (L) limits of maximum pressure in pounds per square inch shall be 2,300 and 1,750, respectively.

3.8.2 **Time from start of pressure rise to maximum pressure.** The time from start of pressure rise to maximum pressure shall be not greater than 0.05 second, maximum.

3.8.3 **Misfire.** There shall be no misfires (see 4.6.6.2).

3.9 **Change in propellant lot.** Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.8. No cartridge firing shall produce results outside the limits of 3.8.1 and 3.8.2.

3.10 **Workmanship.** Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, missing or damaged O-ring, perforated closure disc, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture. All cartridge marking shall be verified as proper (see 4.6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 **Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 **First article inspection.** First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.2 through 3.7.2 and 3.10 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the intended sources of supply.

4.4.1 **Sample size.** The first article sample shall consist of 76 cartridges.

4.4.2 **Inspection routine.** Sixty-six cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests as prescribed in MIL-D-21625: 40-foot drop test (6); temperature and humidity cycling (12); 15 "g" shock (6); vibration (6); −65°F temperature (12); +160°F temperature (12); +70°F temperature (12). The number in parentheses denotes the number of cartridges allocated for each test. No cartridge firings shall produce results outside the limits specified in 3.8. Firings shall be conducted in the test fixture specified in 4.6.6.

4.4.3 **Failure.** Failure of any cartridge shall be cause for rejection of the first article sample.

4.5 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table I.

4.5.1 **Lot size.** Unless otherwise specified (see 6.2), the production lot size shall be not less than 1301 minimum and not greater than 8000 maximum. Cartridges required for all test purposes are selected from the production lot and will not be applied as part of the quantity specified for delivery by the contract. Only primary primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting agency.

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4.5.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II and the provisions of 4.6.6. Unless otherwise specified in the contract, order or requisition, all samples shall be provided by the contractor (see 6.2).

4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement para.	Test para.	Classification (WR-43)
Primary component	Single lot verification	3.4	4.5.1	(M101)
Ignition element	Age and acceptance verification	3.5	4.6.1	(C1)
Ignition element	X-ray	3.5	4.6.5.2	(C2)
Propellant charge	Type and weight	3.6.1	-	(C3)
Smokeless powder	Condition	3.6.3	4.6.2	(C4)
Cartridge	Visual examination	3.7.3	4.6.3	(M102)
Cartridge	Bridge circuit resistance	3.7.1	4.6.4	(C5)
Cartridge	X-ray	3.7.2	4.6.5.1	(C6)
Cartridge	Ballistics	3.8	4.6.6	(C7)
Cartridge	Ballistics (change in propellant lot)	3.9	4.8	(C8)
Cartridge	Preservation-packaging, packing and marking	Section 5	4.6.3	(M103)

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
66 - 110	15	10
111 - 180	15	10
181 - 300	20	10
301 - 500	25	10
501 - 800	30	10
801 - 1300	35	10
1301 - 3200	40	10
3201 - 8000	50	10
8001 - 22000	75	10

4.6 Tests.

4.6.1 Ignition element verification. Verification shall be provided that the ignition elements used were certified and accepted in accordance with the requirements of 3.5.

4.6.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

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4.6.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.7.3 shall be rejected. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5.

4.6.4 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.7.1 shall be ascer tained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to not greater than 25 milliamperes, maximum. The test circuit shall be connected to the cartridge through a fully assembled electrical connector, MS3106-12-5S. A new connector shall be used after a maximum of 500 connections. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.7.1 or fail to assemble completely, by hand, with the specified connector shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.6.5 Radiographic examination.

4.6.5.1 Cartridge. All cartridges shall be radiographically examined in accordance with MIL-STD-453. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.6.5.2 Ignition element. All ignition elements shall be radiographically examined in accordance with MIL-STD-453. Each ignition element shall be mounted in suitable racks or frames and oriented for the most revealing exposure. Pressure sensitive tape may be used as an aid in positioning the ignition element for X-ray. Reject elements that lack styphnate, FA-878, mica washer insulator, or for any other defects in assembly.

4.6.6 Ballistic tests. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for 24 hours at 70°F ± 5°F and fired in a closed cylindrical chamber having an internal volume, excluding cartridge volume, of 2.70 cubic inches, the ratio L/D = 1.0 (approximately), and having provisions for the measurement of pressure. The chambers shall be thoroughly cleaned before each firing. A record of pressure versus time and the time from closing of the firing key to start of pressure rise shall be taken for each firing. Firing current shall be not less than 10 amperes from a 20-volt dc source. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes. Acceptance criteria (regardless of lot size) shall be: accept zero defects and reject if one or more defects.

4.6.6.1 Criteria of acceptability. The upper (U) and lower (L) limits of maximum pressure shall meet the requirements of 3.8.1. The time from start or pressure rise to maximum pressure shall meet the requirement of 3.8.2. A failure of any cartridge to meet these requirements shall be cause for rejection of the lot represented.

4.6.6.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

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4.7 **Retest.** There shall be no retests.

4.7.1 **Test failure.** If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

4.8 **Change in propellant lot.** If the production lot being manufactured is the first production lot produced by the supplier or whenever a change in propellant lot occurs, test firings, in addition to the ballistic test of 4.6.6, shall be conducted as follows: Twelve rounds each shall be fired at −65°F, 70°F, and 160°F. Results of these firings shall fall within the limits of 3.8.

5. PACKAGING

5.1 **Preservation and packaging.** For definitions of packaging levels, see 6.4.

5.1.1 **Level A.** Level A packaging shall be used for packaging of all production lots for service use. Two cartridges shall be packaged in an inner container in accordance with DL491834.

5.1.2 **Level B.** Not applicable.

5.1.3 **Level C.** When specified in the contract (see 6.2), those cartridges which are intended for expenditure in first article sample testing and production lot acceptance testing shall be packaged in accordance with method III of MIL-P-116 and with 49 CFR 171-178.

5.2 **Packing.** For definitions of packing levels, see 6.4.

5.2.1 **Level A.** Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with DL491835.

5.2.2 **Level B.** Not applicable.

5.2.3 **Level C.** When specified in the contract (see 6.2), cartridges intended for expenditure in first article testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with 49 CFR 171-178.

5.3 **Marking.**

5.3.1 **Special marking.** Marking of exterior containers shall be in accordance with 49 CFR 171-178.

5.3.2 **Normal marking.** Unless otherwise specified in the contract or order (see 6.2), the marking information on unit packages and shipping containers shall be as specified on DL491834 and DL491835, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 **Intended use.** The Mark 17 Mod 1 impulse cartridge is intended for use as the power source in stores release ejection systems on aircraft.

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6.2 **Ordering data.** Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required, and if so, specify the test activity
c. Date and place of delivery
d. Inspection conditions, if other than as specified (see 4.3)
e. That revision "C" of MIL-D-21625 be used (see 4.4.2)
f. Lot size, if other than as specified (see 4.5.1)
g. Sampling, if other than as specified (see 4.5.2)
h. Level C packaging requirements (see 5.1.3 and 5.2.3)
i. Normal marking, if other than as specified (see 5.3.2)
j. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.3 **First article.** When a first article inspection is required, the item should be a first article sample. The first article should consist of 84 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Definitions.** For the purpose of this specification, the following definitions apply.

6.4.1 **Level A packaging.** The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.2 **Level C packaging.** The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.4.3 **Level A packing.** The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.4 **Level C packing.** The degree of packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 **Changes from previous issue.** Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-22079C(AS)	CARTRIDGE, IMPULSE, MARK 17 MOD 1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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